F18CF – Linear Algebra

Tutorial Sheet 10: Finite Markov Chains

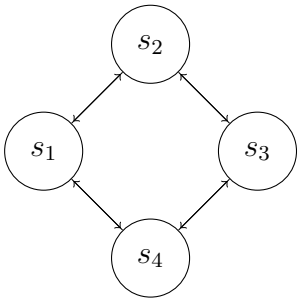
Getting started.

1. (i) This stochastic matrix is regular. One can draw the associated Markov diagram and convince oneself that one can go from any state to any other with an directed path of length 3. Alternatively, one can check that

$$\begin{pmatrix} 1/2 & 1/3 & 1/2 \\ 1/2 & 1/3 & 0 \\ 0 & 1/3 & 1/2 \end{pmatrix}^2 = \begin{pmatrix} 5/12 & 4/9 & 1/2 \\ 5/12 & 5/18 & 1/4 \\ 1/6 & 5/18 & 1/4 \end{pmatrix},$$

which is positive, hence A is regular.

(ii) This stochastic matrix is not regular, one can draw the associated Markov diagram (with the state s_j corresponding to the j-th column of A), where the probabilities are omitted:



From this, we see that a directed path of even length joins states j, j' with the same parity (both odd or both even), while a directed path of odd length joins to states with different parity (one even and one odd). In particular, it is not possible to connect two arbitrary states with directed paths of length n for a fixed integer n, hence A is not regular by the characterisation of regularity from the course.

(iii) This stochastic matrix is regular. One can draw the associated Markov diagram and convince oneself that one can go from any state to any other with an directed path of length 3. Alternatively, one can check that

$$\left(3\begin{pmatrix} 1/3 & 1/3 & 0 & 0 & 0 & 1/3 \\ 1/3 & 1/3 & 1/3 & 0 & 0 & 0 \\ 0 & 1/3 & 1/3 & 1/3 & 0 & 0 \\ 0 & 0 & 1/3 & 1/3 & 1/3 & 0 \\ 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 1/3 & 0 & 0 & 0 & 1/3 & 1/3 \end{pmatrix}\right)^3 = \begin{pmatrix} 7 & 6 & 3 & 2 & 3 & 6 \\ 6 & 7 & 6 & 3 & 2 & 3 \\ 3 & 6 & 7 & 6 & 3 & 2 \\ 2 & 3 & 6 & 7 & 6 & 3 \\ 3 & 2 & 3 & 6 & 7 & 6 \\ 6 & 3 & 2 & 3 & 6 & 7 \end{pmatrix},$$

which is positive, hence A is regular. (Note that we used a factor 3 at the start so that the entries of the matrix become 1, making the computations much easier - and this has of course no effect on whether our matrix is regular.)

2. The general districbution corresponds to an eigenvector for the eigenvalue 1, hence a vector solution of the system $(A - I_n)\vec{x} = \vec{0}$.

   (i) We solve the corresponding system via Gaussian elimination:

$$\left(\begin{array}{cc|c} -1/2 & 1/4 & 0 \\ 1/2 & -1/4 & 0 \end{array}\right) \rightsquigarrow \left(\begin{array}{cc|c} -1/2 & 1/4 & 0 \\ 0 & 0 & 0 \end{array}\right),$$

   hence there is one free variable (which we knew from Perron-Frobenius) $y = \alpha$, and by susbstitution we get $x = \alpha/2$. Thus, the stationary distribution is of the form

$$\begin{pmatrix} \alpha \\ 2\alpha \end{pmatrix}.$$

   (ii) We solve the corresponding system via Gaussian elimination:

$$\left(\begin{array}{ccc|c} -1/2 & 1/4 & 1/2 & 0 \\ 1/2 & -1/2 & 0 & 0 \\ 0 & 1/4 & -1/2 & 0 \end{array}\right) \rightsquigarrow \left(\begin{array}{ccc|c} -1/2 & 1/4 & 1/2 & 0 \\ 0 & -1/4 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right),$$

   hence there is one free variable (which we knew from Perron-Frobenius) $z = \alpha$, and by susbstitution we get $x = y = 2\alpha$. Thus, the stationary distribution is of the form

$$\begin{pmatrix} 2\alpha \\ 2\alpha \\ \alpha \end{pmatrix}.$$

Problems.

3. **Migration of population.**

   (a) The matrix A is stochastic and positive, hence regular, so Perron-Frobenius applies, and in particular the sequence $\begin{pmatrix} c_n \\ r_n \end{pmatrix} = A^n \begin{pmatrix} c_0 \\ r_0 \end{pmatrix}$ converges towards a stable distribution. Thus, both sequences c_n and r_n converge.

   (b) The general form of the stationary distribution is obtained by finding the general form of an eigenvector for the eigenvalue 1, which we find by solving the system $(A - I_2)\vec{x} = \vec{0}$ via Gaussian elimination:

$$\left(\begin{array}{cc|c} -0.04 & 0.01 & 0 \\ 0.04 & -0.01 & 0 \end{array}\right) \rightsquigarrow \left(\begin{array}{cc|c} -0.04 & 0.01 & 0 \\ 0 & 0 & 0 \end{array}\right),$$

   hence there is one free variable (which we knew from Perron-Frobenius) $y = \alpha$, and by susbstitution we get $x = \alpha/4$. Thus, the stationary distribution is of the form

$$\begin{pmatrix} \alpha \\ 4\alpha \end{pmatrix}.$$

Thus, at equilibrium, the total poluation is $\alpha+4\alpha = 5\alpha$ when the population living inside the city is α, hence

$$\frac{\alpha}{5\alpha} = 20\%$$

of the population will be living in that city (note that we did not need to know the total number of people living in the country).

(c) We could compute the characteristic polynomial of A and find its roots as usual (you should do it, to double-check you still know how to do it without mistake). However, there is a faster method: We know that $\lambda_1 = 1$ is an eigenvalue since A is stochastic, and we can find the second eigenvalue λ_2 using the trace, since by a result of the course:

$$\lambda_1 + \lambda_2 = \mathrm{tr}(A) = 1.95,$$

hence $\lambda_2 = 0.95$. A is a 2×2 matrix with two distinct eigenvalues, hence A is diagonalisable.

(d) Since A is diagonalisable, there exists a basis $\vec{e}_1, \vec{e}_2$ of $\mathbb{R}^2$ made of eigenvectors of A. Let us write the initial vector $(c_0, r_0)^T$ in this basis:

$$\begin{pmatrix} c_0 \\ r_0 \end{pmatrix} = a_1\vec{e}_1 + a_2\vec{e}_2$$

for some constants $a_1, a_2 \in \mathbb{R}$. We thus have for $n \geq 1$:

$$\begin{pmatrix} c_n \\ r_n \end{pmatrix} = A^n \begin{pmatrix} c_0 \\ r_0 \end{pmatrix} = a_1 A^n \vec{e}_1 + a_2 A^n \vec{e}_2 = a_1\lambda_1^n\vec{e}_1 + a_2\lambda_2^n\vec{e}_2.$$

In particular, we see by looking at the second component of that vector that c_n is of the form $\alpha_1\lambda_1^n + \alpha_2\lambda_2^n$, where $\alpha_1, \alpha_2 \in \mathbb{R}$ are constants that only depend on $a_1, a_2, \vec{e}_1, \vec{e}_2$.

(e) We find the constants α_1, α_2 by evaluating at $n = 0$ and $n = 1$, since we know that $c_0 = 10^6$ and $c_1 = 0.96 \times 10^6 + 0.01 \times 10^7 = 1.06 \times 10^6$ which yields the system

$$\begin{cases} \alpha_1 + \alpha_2 &= 10^6 \\ \alpha_1 + 0.95\alpha_2 &= 1.06 \times 10^6 \end{cases},$$

which we solve as usual via Gaussian elimination, which yields $\alpha_1 = 2.2 \times 10^6$, $\alpha_2 = -1.2 \times 10^6$.

In particular, we have $c_n = 2.2 \times 10^6 - 1.2 \times 10^6 (0.95)^n \to_\infty 2.2 \times 10^6$, hence $c_\infty = 2.2 \times 10^6$, and we have

$$|c_n - c_\infty| = 1.2 \times 10^6 \times (0.95)^n \leq \frac{1}{100} 2.2 \times 10^6$$

if and only if

$$(0.95)^n \leq 11/600$$

if and only if

$$n \geq \frac{\log(11/600)}{\log(0.95)} \simeq 77.96$$

We will thus have $|c_n - c_\infty| \leq \frac{1}{100} c_\infty$ after 78 years.

4. (a) The transition matrix associated to this Markov diagram is:

$$\begin{pmatrix} 0 & 0 & 0 & 1/5 & 1/3 \\ 2/5 & 0 & 1/2 & 1/5 & 1/3 \\ 2/5 & 1/2 & 0 & 2/5 & 1/6 \\ 1/5 & 1/2 & 0 & 0 & 1/6 \\ 0 & 0 & 1/2 & 1/5 & 0 \end{pmatrix},$$

and we check with a computer that

$$\begin{pmatrix} 0 & 0 & 0 & 1/5 & 1/3 \\ 2/5 & 0 & 1/2 & 1/5 & 1/3 \\ 2/5 & 1/2 & 0 & 2/5 & 1/6 \\ 1/5 & 1/2 & 0 & 0 & 1/6 \\ 0 & 0 & 1/2 & 1/5 & 0 \end{pmatrix}^3 = \begin{pmatrix} 3/25 & 7/60 & 1/15 & 38/375 & 77/900 \\ 69/250 & 77/300 & 3/10 & 88/375 & 127/450 \\ 32/125 & 41/150 & 17/60 & 227/750 & 113/450 \\ 21/125 & 19/75 & 7/60 & 11/50 & 34/225 \\ 9/50 & 1/10 & 7/30 & 53/750 & 23/100 \end{pmatrix},$$

which is positive, hence A is regular.

(b) Adapting the PageRank algorithm (with a damping factor of zero), we obtain the ranking vector by computing an eigenvector for the eigenvalue 1, which we obtain by solving the system $(A - I_5)\vec{x} = \vec{0}$ via Gaussian elimination (with a computer, because there is a limit to our mathematical masochism):

$$\left(\begin{array}{ccccc|c} -1 & 0 & 0 & 1/5 & 1/3 & 0 \\ 2/5 & -1 & 1/2 & 1/5 & 1/3 & 0 \\ 2/5 & 1/2 & -1 & 2/5 & 1/6 & 0 \\ 1/5 & 1/2 & 0 & -1 & 1/6 & 0 \\ 0 & 0 & 1/2 & 1/5 & -1 & 0 \end{array}\right),$$

and we find that a basis for the associated eigenspace is the (ranking) vector

$$\begin{pmatrix} 75 \\ 214 \\ 218 \\ 145 \\ 138 \end{pmatrix}.$$

In particular, it follows from the PageRank algorithm that Team 3 wins this tournament, and the podium is: Team 3 (1st), Team 2 (2nd), Team 4 (3rd).